Formal Review of the Latest v2.1 Responses

Remaining gaps in the proposed split-forest decidability proof for $\mathcal{ALCI}_{\mathsf{RCC5}}$

Review memorandum

May 2026

Abstract

This memorandum reviews the latest submitted responses, `completeness_extraction_ALCIRCC5_v2(1).tex` and `sibling_interface_descriptors_ALCIRCC5_v2(1).tex`, together with their PDFs. The update is a real improvement: the density/Hasse objection is now treated in the right way via witness-generated submodels and generated cover presentations; EQ-modalities are normalized under strong equality; and the documents try to separate blocking from semantic EQ. However, the proof still does not establish decidability as written. Several of the claimed v2.1 repairs introduce new formal problems. The most serious are: a contradiction between profile-level equality ports, self-loop blocking, vertical separation, and mosaic reflexivity; an invalid finite-index argument for equality ports; a still-unresolved multiple-upward-PP-witness problem; an incorrect assumption that equality mates have a unique semantic parent; a canonical-unfolding/projection definition that is not compatible with the rootless containment orientation; and an overstrong finite-mosaic arity claim.

1 Executive assessment

The latest response should be viewed as a useful draft of a repair, not as a closed proof. It correctly moves the proof away from arbitrary semantic Hasse diagrams and toward the intended model class: sparse witness-generated abstract models with a generated containment presentation. This substantially improves the earlier proof architecture.

Nevertheless, I would not yet rely on the latest v2.1 documents as a proof of $\mathcal{ALCI}_{\mathsf{RCC5}}$ decidability. The central issue is not the old density objection; that issue is now essentially resolved. The remaining issues are internal to the finite certificate itself, especially the interaction between:

blocking/profile repetition, semantic EQ quotienting, multiple generated superpart witnesses.

The current equality-port repair is inconsistent with the self-loop certificate needed for the motivating formula

$$C_{\uparrow} = \exists\mathsf{PP}.\top \sqcap \forall\mathsf{PP}.\exists\mathsf{PP}.\top.$$

Also, the current parent-function treatment cannot yet support formulas whose satisfaction requires several incomparable proper-superpart witnesses for the same semantic element.

2 What has improved

The following changes are positive and should be retained.

2.1 Witness-generated submodels

The witness-generated submodel lemma in `completeness_extraction_ALCIRCC5_v2(1).tex`, lines 430–512, is the right conservativity step. It reduces arbitrary satisfying abstract models to sparse models generated by selected existential witnesses. The closure is now closed under NNF complement, so the converse direction for universal formulas uses the selected witness for $\exists R.\overline{F}$ correctly.

This largely removes the old density/Hasse concern. The proof no longer needs to assume that the original semantic PP/PPI relation has immediate covers.

2.2 Generated covers rather than ambient Hasse covers

The generated-cover reinterpretation in lines 520–570 is also the right conceptual move. A tree edge should be a generated witness-cover edge relative to the selected presentation, not a Hasse edge of an arbitrary ambient PP-order.

2.3 Elimination of EQ modalities under strong equality

The normalization lemma in lines 302–358 correctly observes that, under strong EQ,

$$\exists \mathsf{EQ}.D \equiv D, \qquad \forall \mathsf{EQ}.D \equiv D.$$

This is a good simplification. After this preprocessing, the modal roles that need witness handling are only $\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}$.

2.4 Recognition that blocking is not semantic equality

The v2.1 text explicitly tries to distinguish finite blocking from semantic EQ, for example around lines 752–767 and 1196–1381. This distinction is essential. Unfortunately, the particular equality-port implementation is still not coherent, as explained next.

3 Critical remaining gaps

3.1 G1. Profile-level equality ports contradict self-loop blocking

This is the most serious defect in the current v2.1 repair.

The finite quotient explicitly allows a parent self-loop $\mathsf{par}_Q(\sigma) = \sigma$ to represent an infinite upward PP-chain; see lines 752–767. The reachability definition then says that a self-loop implies $\sigma \rightsquigarrow_{\mathsf{PP}} \sigma$; see lines 1718–1724.

The new vertical-separation axiom says that if $\sigma \rightsquigarrow_{\mathsf{PP}} \sigma'$ or $\sigma \rightsquigarrow_{\mathsf{PPI}} \sigma'$ strictly, then no profile-port pair (σ, p) may be $\equiv_{\mathsf{EQ},Q}$-related to (σ', q); see lines 1274–1285. Therefore, for a self-loop it entails

$$(\sigma, p) \not\equiv_{\mathsf{EQ},Q} (\sigma, p)$$

for every port p. The proof of Lemma 10.5 explicitly uses this conclusion at lines 1326–1332.

But mosaics also require reflexive equality. Axiom (M1) says $\mu(v, v) = \mathsf{EQ}$ for every vertex; see lines 843–849. The revised axiom (M7) says

$$\mu(v, w) = \mathsf{EQ} \quad \Longleftrightarrow \quad v \equiv^{\mu}_{\mathsf{EQ},Q} w$$

for all vertices v, w; see lines 1252–1260. Taking $v = w$, M1 and M7 force

$$(\mathrm{st}(v), \pi(v)) \equiv_{\mathsf{EQ},Q} (\mathrm{st}(v), \pi(v)).$$

For a vertex of self-loop profile σ, this is exactly the relation that vertical separation forbids.

Thus the one-state certificate for $C_{\uparrow}$ is made invalid by the very repair intended to support it. The problem is structural: a single finite relation on profile-port pairs cannot distinguish "the same occurrence" from "the next lap of the blocking cycle" when both carry the same profile and the same port tuple.

Required repair. The equality mechanism must be occurrence-sensitive or context-sensitive. One possible direction is:

- treat $v = w$ in mosaics as identity, not as a global $\equiv_{\mathsf{EQ},Q}$ lookup;
- use finite equality variables local to a bounded mosaic/context, not a global relation on reusable profile-port pairs; and
- state explicitly that equality links are between distinct displayed occurrences in a finite context, while blocking repetition never induces such a link across cycle laps.

As written, the current global profile-port relation $\equiv_{\mathsf{EQ},Q}$ cannot serve both roles.

3.2 G2. The finite equality-port bound is not proved and appears false

Lines 1348–1355 say that equality ports are drawn from a finite alphabet whose size is bounded by the number of distinct semantic elements that can appear in a witness-generated submodel cell, and hence by a computable function of $|\,\mathrm{cl}(C_0)|$. Lines 1383–1408 then define the extracted congruence by using the origin element itself as the canonical identity index.

This does not yield a finite alphabet. A witness-generated model may contain infinitely many distinct semantic elements of the same rank-d profile. In strong EQ semantics, each distinct element is its own equality class. If one port is needed per origin to avoid false equality, the port alphabet is infinite. If finitely many ports are reused for distinct origins, then M7 may force false EQ labels when two reused ports appear together in a mosaic.

The sentence in lines 1405–1408 only bounds the number of ports contributed by one semantic element; it does not bound the number of semantic elements. Hence Lemma 10.8 does not establish the finite-index property.

Required repair. Do not try to encode global semantic identity by finite reusable profile ports. Instead, equality should be stored as bounded local matching data for the finite interfaces that actually appear in a certificate. If global equality classes are needed, a separate finite abstraction theorem must prove that only boundedly many equality representatives per profile are ever relevant.

3.3 G3. Multiple upward PP witnesses are still not handled

The containment orientation makes $\exists\mathsf{PP}.D$ point upward to a proper superpart. A single occurrence has only one parent. However, a semantic element may need several incomparable proper-superpart witnesses.

The split-tree construction in lines 595–604 assigns a single parent to an occurrence. For each $\exists\mathsf{PP}.D$, it either uses the existing parent or sets $\mathsf{par}(u) = u^{\uparrow}$. This does not define what happens when one occurrence has two incompatible upward witnesses. The later splitting clause, lines 620–624, only says that if a single element b must appear under multiple parents, then one should introduce one occurrence per containment branch. The problematic case is the dual one: the same lower element a may need to appear under several selected superparts.

A representative stress formula is:

$$\begin{aligned} C_{AB} := {} & \exists\mathsf{PP}.A \sqcap \exists\mathsf{PP}.B \\ & \sqcap \forall\mathsf{PP}(A \to (\forall\mathsf{PP}.\neg B \sqcap \forall\mathsf{PPI}.\neg B)) \\ & \sqcap \forall\mathsf{PP}(B \to (\forall\mathsf{PP}.\neg A \sqcap \forall\mathsf{PPI}.\neg A)). \end{aligned}$$

This is satisfiable by an element x with two incomparable proper superparts y, z, with $A(y)$, $B(z)$, and $y\mathsf{PO}z$. A single parent chain above one occurrence of x cannot contain both witnesses, because the formula forbids the A- and B-witnesses from being comparable. The intended split-copy idea can handle this only by creating two occurrences of x, one under y and one under z, and later quotienting them as semantic equality mates.

The current v2.1 text has not formally defined this construction. In fact, Lemma 12.3 goes in the opposite direction: lines 1968–1971 say that equality mates have either no parent or the same unique semantic parent. But the whole reason to split is that equality mates of the same semantic element may need different generated parents.

Required repair. The parent function should be a function on occurrences, not on semantic equality classes. Equality mates must be allowed to have different generated parents. Therefore par_Q should not be required to descend to mate classes. Eventuality discharge should quantify over equality-mate occurrences or over a finite set of parent contexts, without imposing parent congruence.

3.4 G4. The canonical unfolding/projection is not compatible with the rootless orientation

The canonical unfolding definition in lines 1865–1875 says that $U(Q)$ is a rooted tree of finite walks in (S, par_Q) starting at the evaluation profile σ_*, with parent map given by one-step predecessor. The projection $\pi : T \to U(Q)$ in lines 1878–1888 uses the depth of an occurrence relative to a root occurrence u_*.

This is not compatible with the containment-oriented construction. The evaluation occurrence may have generated parents above it; indeed this is necessary for $\exists\mathsf{PP}$-requirements. After upward expansion, u_* is not a root of the containment tree. Ancestors of u_* do not have a finite depth below u_*, and the displayed formula for $\pi(u)$ is not defined for them.

There is also an orientation problem: if $\mathsf{par}_Q(\sigma_i) = \sigma_{i+1}$, then the walk $(\sigma_*, \sigma_1, \ldots, \sigma_m)$ moves upward along the containment parent function. But the parent map of the rooted walk tree is truncation to the predecessor, which reverses the intended semantic parent relation. This may be a notational repair, but as written the projection is not a faithful unfolding map.

The surjectivity claim in Lemma 12.2 is also not justified. Given an arbitrary finite walk in (S, par_Q), choosing any occurrence with the endpoint profile does not ensure that its actual parent chain realizes the specified preceding profiles.

Required repair. Use a pointed bi-oriented unfolding: nodes should be finite paths extending both above and below the evaluation occurrence, or else represent upward trunks as one-hole contexts/lassos rather than rooted walks from σ_*. The projection must be defined on occurrence paths, not merely on endpoint profiles.

3.5 G5. The side-context witness lemma remains circular

The side-context witness lemma in lines 1007–1185 is an improvement in form, but its proof still assumes the placement it is supposed to prove.

The witness-menu extraction in lines 961–1003 says that a DR/PO witness occurrence exists, "possibly as a sibling occurrence under a common ancestor." The side-context lemma then proves existence of such a v by citing Construction 5.1, step 4. But step 4, lines 614–618, merely says to record an outgoing witness edge to a node mapped to b, "typically a sibling of u under a common ancestor." It does not construct the occurrence, the common ancestor, or the corresponding containment placement.

The use of a lowest common ancestor also requires care. In a rootless or upward-regular containment presentation, and especially after split copies, there need not be a unique LCA in the naive rooted-tree sense. The lemma should not use an LCA until the underlying generated presentation has been formally shown to be an actual tree with the relevant nodes in the same component and with a well-defined meeting point.

Finally, a five-vertex side context checks only a bounded immediate interface. It does not by itself prove that all deeper universal restrictions and all composition constraints involving descendants of u or v remain coherent. The text says these are handled by support closure, but the support-closure argument is itself still incomplete; see G6.

Required repair. State and prove a genuine placement theorem: for every selected DR/PO witness, the split presentation can be extended by a finite side context such that all induced vertical, sibling, and equality constraints are preserved. The proof must construct the context rather than cite the schematic step 4.

3.6 G6. The finite mosaic arity theorem is overclaimed

Theorem 9.3, lines 1619–1660, claims that arity 5 suffices for all mosaics, using three ingredients: local arity of axioms (M1)–(M7), local arity of support-closure operations (SC1)–(SC4), and the RCC5 patchwork property.

This does not yet prove the needed result. Checking that each axiom is binary or ternary is not the same as proving that descriptor membership, closure under iterated extensions, and global realizability are determined by all submosaics of size at most 5. In particular:

- SC4 can be iterated along containment chains. The proof says that adding one generated child raises arity by one and stays within arity 5, but an iterated construction may couple several extensions through shared endpoints.
- The patchwork property is an amalgamation principle for compatible networks; it does not automatically imply a finite forbidden-subnetwork theorem of arity 5 for the enriched descriptor language with statuses, equality ports, witness menus, and support closure.
- The theorem is used to justify finite enumeration of support-closed mosaics, but the closure object is defined semantically as containing joint networks occurring in a model. The equivalence between semantic occurrence and bounded syntactic closure is not proved.

Required repair. Either give a precise finite closure algorithm and prove by induction that it covers every finite prefix network needed in the soundness construction, or replace the constant 5 by a computable bound derived from the finite state space and prove a finite-model/compactness bound for descriptors. The current patchwork citation is not sufficient.

3.7 G7. Simultaneous realization still relies on the hidden source model

Lemma 12.4, lines 1978–2099, tries to prove simultaneous realization by taking $U^*(Q) = T$, where T is the split tree extracted from the original model. For the model-to-quotient completeness direction, this can be useful as a witness that the extracted data were realizable. But the decision procedure needs a finite validity condition that guarantees realization for an arbitrary enumerated finite certificate, without access to a hidden model T.

As written, the proof of simultaneous realization repeatedly appeals to the source model $\mathcal{I}_w$, the original witness selector, and the origin map. For example, SR1 uses the existence of a witness in $\mathcal{I}_w$ and then asserts that the split-tree construction installs a finite generated cover chain visiting it. This proves at most that the particular extracted tree worked, assuming the earlier construction was correct. It does not provide a decidable certificate-side criterion independent of the model.

This matters because the main theorem later combines completeness with soundness by enumerating finite quotients. The finite validity check must be strong enough that every accepted quotient has a coherent unfolding. The current argument does not yet bridge that gap, especially given G1–G6.

Required repair. Define validity purely on finite data and prove a soundness theorem from that finite validity condition to an unfolded model. If reachability is the chosen replacement for lasso/Buechi acceptance, then the proof must show that finite reachability plus the finite mosaic/equality conditions entail one simultaneous unfolding for all eventualities, not merely for extracted quotients.

3.8 G8. The sibling-interface note is out of sync with v2.1

The sibling-interface document remains useful as a high-level compatibility note, but it does not yet match the latest v2.1 completeness manuscript.

First, lines 459–465 state a simple reachability-discharge condition from σ alone. The v2.1 completeness manuscript changed this to mate-class-aware reachability. The sibling note therefore still states the old v2 condition, not the claimed v2.1 condition.

Second, lines 265–268 allow a literal parent self-loop in the generated split-tree presentation. A literal occurrence-level self-loop would violate irreflexivity of PP/PPI. Self-loops should exist only in the finite quotient/blocking graph, whose unfolding creates fresh occurrences at each lap. The sibling note says this later, but the definition itself remains misleading.

Third, the sibling note claims the v1 soundness chain is unchanged, while the v2.1 completeness paper substantially changes equality handling. If typed EQ, M7, equality ports, and vertical separation are part of quotient validity, then the soundness chain must be restated against those actual v2.1 validity conditions.

Required repair. Update the sibling-interface note to v2.1 terminology. In particular, remove literal occurrence-level self-loops, replace simple reachability by the final correct equality-aware eventuality condition, and restate the soundness theorem for the actual equality mechanism used by the completeness proof.

4 Additional technical concerns

4.1 The open-cross-edge lemma is still too compressed

The open-cross-edge lemma in lines 808–825 of the completeness manuscript is a key structural claim: non-core descendants of sibling subtrees can have only DR or PO cross-relations. The proof is a short appeal to the old open-cross-edge theorem. But after generated covers, split copies, equality ports, and mate-aware reachability are introduced, the notion of core must be rechecked. In particular, if equality mates may appear under different parents, then whether a node is “visible to both sides” can depend on the chosen occurrence context. The old lemma may still be true, but it needs to be restated and proved under the new occurrence/equality semantics.

4.2 The RCC5 composition table is no longer displayed

The latest completeness manuscript now cites a standard RCC5 composition table instead of printing the earlier incorrect one. This avoids the previous error, but it also makes the paper less self-contained. Since the proof relies on composition at several critical points, the final manuscript should include the correct atomic RCC5 table, or at least cite a fixed external source and state which convention is used.

4.3 The phrase "all six obligations discharged" is too strong

The v2.1 changelog says that all six review obligations are discharged. Given G1–G8, that statement is not yet supportable. A safer statement would be:

> The present revision proposes repairs for the six obligations identified in the v2 review. Several repairs, especially the explicit equality-port mechanism, the multiple-parent split construction, and the finite mosaic arity bound, still require further formal verification.

5 Recommended path to a correct proof

The current direction is promising, but the certificate needs a different formalization of equality and upward witnesses. I recommend the following repair sequence.

5.1 Step 1: Separate three relations completely

The proof should maintain three distinct notions:

$$\begin{aligned} \text{blocking repetition} &: \text{ same finite profile in the quotient,} \\ \text{occurrence identity} &: \text{ the same vertex in a finite unfolding/context,} \\ \text{semantic } \mathsf{EQ} &: \text{ a declared typed congruence between displayed occurrences.} \end{aligned}$$

A reusable profile-port pair cannot by itself encode occurrence identity. M7 should not require global $\equiv_{\mathsf{EQ},Q}$-reflexivity on profile-port pairs when those pairs are reused across blocking laps.

5.2 Step 2: Make split copies explicit for upward witnesses

For each occurrence and each upward PP-eventuality, the construction should be allowed to create a fresh occurrence-copy of the lower element under the chosen superpart. These copies may later be semantically quotiented, but their parents need not agree. Therefore parent preservation must not be an axiom of semantic EQ.

A finite certificate can store this as a finite set of parent contexts rather than a single parent function per semantic equality class.

5.3 Step 3: Redefine the unfolding as a generated occurrence grammar

Instead of rooted walks from σ_*, use a finite occurrence grammar or context graph whose generated objects are occurrence paths around the evaluation point. Blocking cycles in this grammar create fresh occurrences. Equality links are local edges between generated occurrences, subject to typed-congruence checks, not profile-level identities.

5.4 Step 4: Prove finite descriptor locality independently

The support-closed mosaic construction should be given as an explicit finite closure algorithm. The proof should show, by induction on finite prefixes, that every required finite prefix network is represented by the finite descriptor language. If a constant arity bound is claimed, prove it directly; otherwise use a larger but computable bound.

5.5 Step 5: Re-sync the sibling-interface and completeness papers

Once the equality and parent-context formalism is fixed, the sibling-interface note should be rewritten to match the final definitions exactly. It should not refer to v1 soundness as unchanged unless every altered v2.1 validity condition has been checked against the v1 soundness proof.

6 Conclusion

The latest response is materially better than the earlier v2 draft. It correctly removes the density/Hasse obstruction and adopts the intended sparse, witness-generated model class. However, the decidability proof still does not stand as written.

The decisive remaining problem is the finite certificate's equality and parent structure. The new $\equiv_{\mathsf{EQ},Q}$ mechanism simultaneously needs to be reflexive for mosaic self-edges and non-reflexive across self-loop blocking laps; a global relation on profile-port pairs cannot do both. Moreover, equality mates must be allowed to have different generated parents in order to realize multiple incomparable upward PP-witnesses, whereas the current proof assumes a unique semantic parent.

Thus the current v2.1 manuscripts should be treated as a useful repair proposal, not as a complete proof. The proof can likely be repaired, but it needs a cleaner occurrence-level equality formalism and an explicit parent-context construction for split copies before the finite quotient theorem can be accepted.